

The UNIVERSITY OF AZAD JAMMU & KASHMIR

DEPARTMENT OF SOFTWARE ENGINEERING

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Domain Model

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FOOD DELIVERY SYSTEM

Software Design & Architecture Group Activity (Tasks 1–4)

Task 1: Candidate Conceptual Classes

We use both the noun-phrase technique and the category-list technique to identify candidate classes in the Food Delivery System.

No.	Candidate class	Purpose
1	**Customer**	Person who places food orders.
2	**Restaurant**	Business that prepares food.
3	**Menu**	Collection of food items offered by a restaurant.
4	**MenuItem**	Individual food item with a price.
5	**Order**	A customer's request for food.
6	**OrderItem**	A menu item and quantity included in an order.
7	**DeliveryDriver**	Person who delivers an order.
8	**Payment**	Records payment for an order.
9	**Rating**	Customer feedback or score.
10	**Address**	Delivery location saved for a customer.

Noun-phrase technique: Customer, Restaurant, Menu, MenuItem, Order, OrderItem, DeliveryDriver, Payment, Rating, and Address.

Category-list technique: People (Customer, DeliveryDriver), business (Restaurant), products (MenuItem), collection (Menu), transactions (Order, Payment), transaction details (OrderItem), feedback (Rating), and location (Address).

Task 2: UML Class Diagram

The diagram includes all 10 candidate classes. Each class separates attributes (state/data) from methods (behavior/actions).



The three important classes selected for detailed explanation are Customer, Order, and DeliveryDriver.

Three Important Classes: Attributes and Methods

1. Customer

Attributes (State)	Methods (Behavior)
customerId: int	register()
name: String	placeOrder()
phone: String	giveRating()
address: String	

2. Order

Attributes (State)	Methods (Behavior)
orderId: int	calculateTotal()
orderDate: Date	confirmOrder()
status: String	cancelOrder()
totalAmount: double	updateStatus()

3. DeliveryDriver

Attributes (State)	Methods (Behavior)
driverId: int	acceptDelivery()
name: String	pickUpOrder()
phone: String	deliverOrder()
availability: String	updateAvailability()

Task 3: Abstraction Decisions

Abstraction means showing only the important information and hiding unnecessary real-world details. In our Food Delivery System, we store a customer's name, phone number, and address, but we do not need to store their height, eye colour, or clothing details. For a delivery driver, we store their availability and delivery status, but we ignore personal details that are not relevant to delivering orders. This makes the system simpler, easier to manage, and more efficient.

Task 4: One Invented Software Object

Invented object: OrderValidator

An OrderValidator is a software object that checks whether an order is valid before it is submitted. It verifies that the customer has selected at least one menu item, the items are available, and the order contains the required information. If something is missing or invalid, it displays an error message. This object is invented for software purposes because it is not a physical object in the real world.